

EES203: Geochemistry

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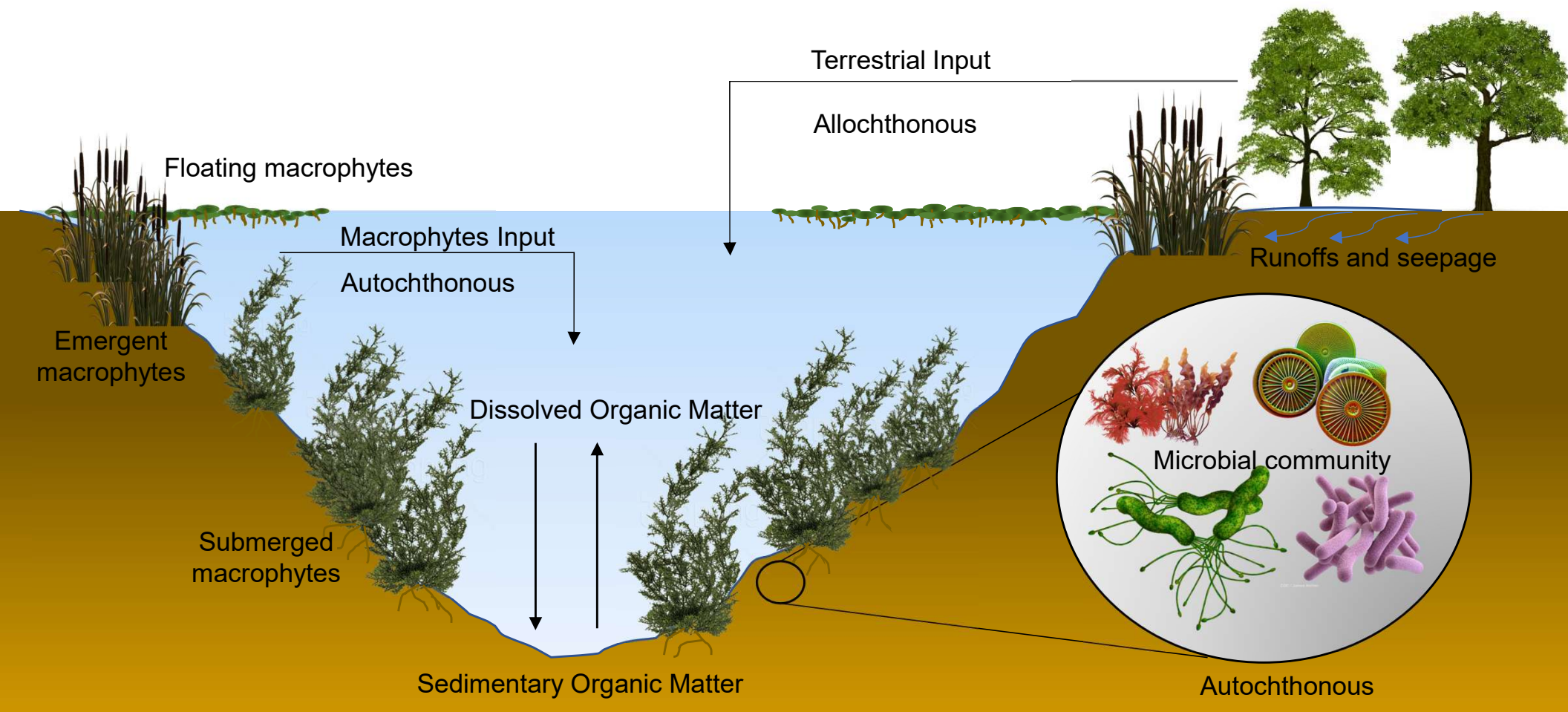
Organic geochemistry

Depositional environment associated with accumulation of organic matter

1. Lacustrine environments

- Closed and open lakes
- Autochthonous and allochthonous OM sources

OM sources



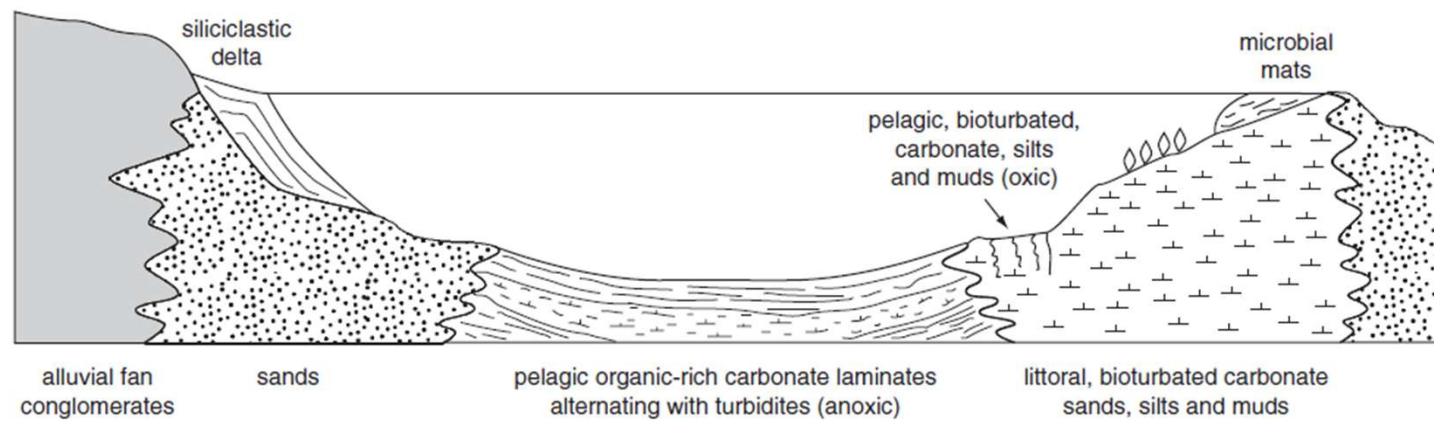


Fig. 3.27 Generalized depositional components in a hydrologically open, freshwater lake (after Eugster & Kelts 1983).

Composition of organic matter

- Aquatic organic matter from phytoplankton is rich in N due to its high protein and lipid content, and has low C/N ratios (commonly between 4 and 10).
- Terrestrial organic matter is dominated by fibrous tissues, cellulose and lignin, which are N-poor, and have C/N ratios of 20 or higher. The C/N ratio is here used to distinguish changes in aquatic and terrestrial organic matter sources in the lake (Meyers, 2003).
- In peatlands, the C/N ratio is an indicator of may indicate changes in the type of peat-forming plants (Kuhry and Vitt, 1996).

Organic matter source identification: Lacustrine and oceanic setting



Composition of C/N in modern lake system

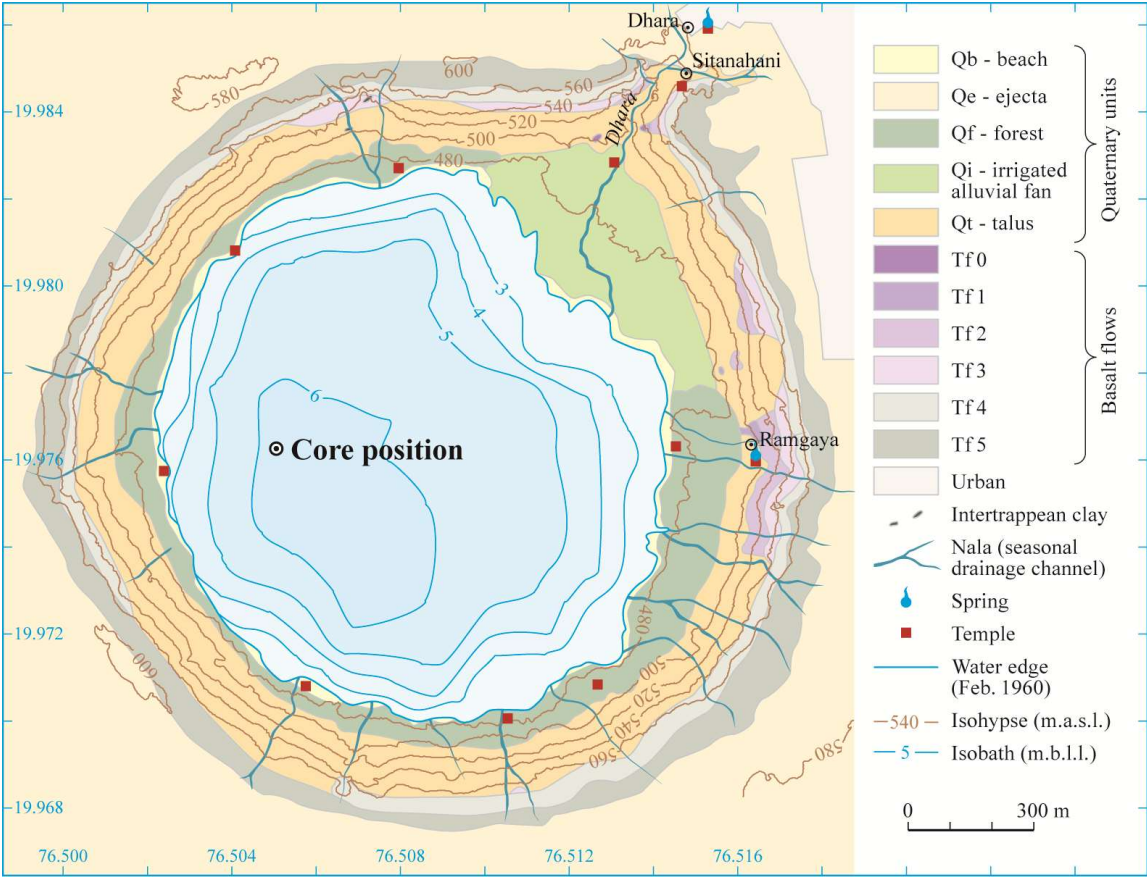
Sample label	Organic matter source	C/N
A	Macroalgae	14.88
T	Mixed plankton	10.86
H1	Herbaceous	24.68
H2	Herbaceous	25.82
H3	Herbaceous	19.49
L1	Lichen	63.55
L2	Lichen	65.73
S	Surface soil	16.99
VV	Animal excrement	21.54

Lake Meidsee (C/N content)

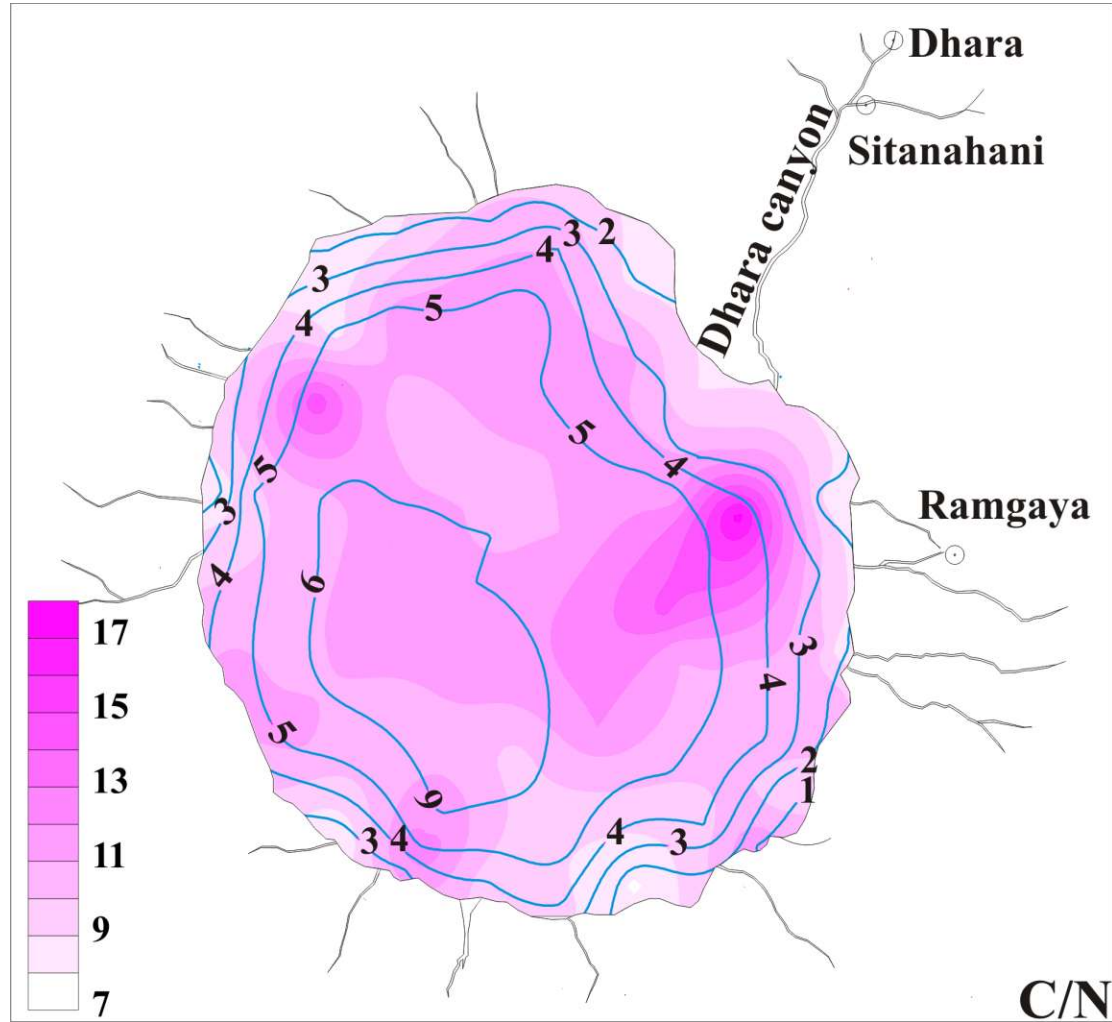
(Thevnon et al., 2012)

C/N ratios in a sediment core from Lonar lake, central India —usefulness as an organic source indicator

Lonar lake, Central India

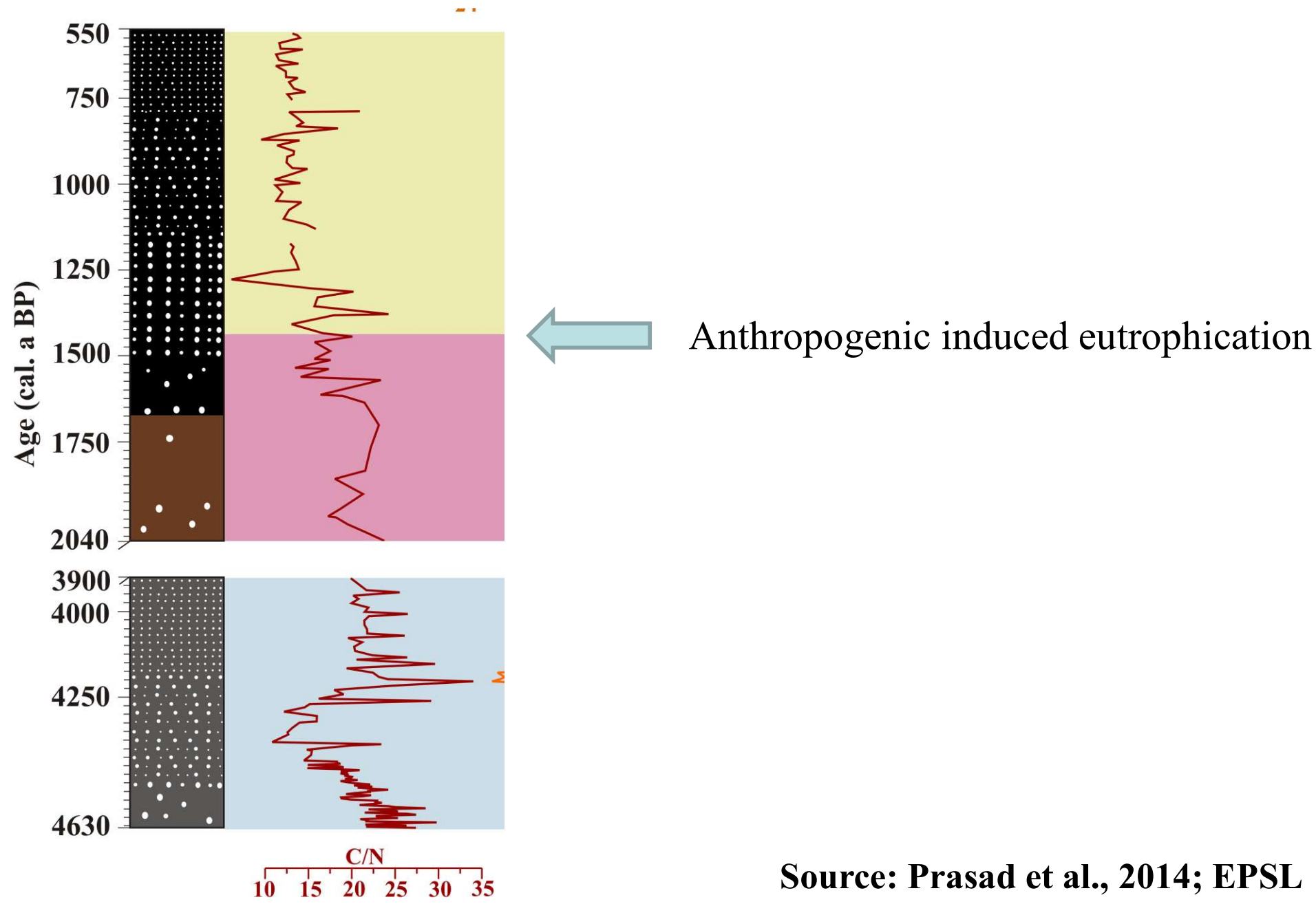


Modern C/N values in the Lonar lake



Source: Basavaiah et al., 2014; FAL

**C/N ratios in a sediment core from Lonar lake, central India —usefulness
as an organic source indicator**



Source: Prasad et al., 2014; EPSL

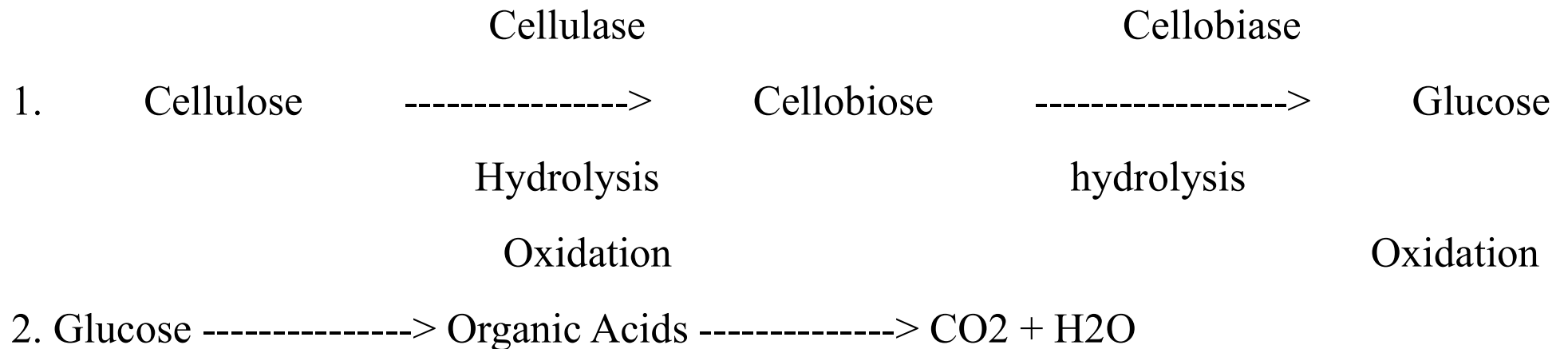
Microbial degradation of organic matter during diagenesis

- The composition of the organic matter preserved depends primarily by the heterotrophic microorganisms
- Amino acids, peptides and carbohydrates are soluble and assimilated directly by the decomposer community
- Insoluble materials (proteins and polysaccharides) can be hydrolysed by extracellular enzymes of fungi to their water soluble constituents aminoacids and monosaccharides
- Rate of decomposition differs for various compound classes

Degradation of bipolymers

Cellulose decomposition

The decomposition of cellulose occurs in two stages: (i) in the first stage the long chain of cellulose is broken down into cellobiose and then into glucose by the process of hydrolysis in the presence of enzymes cellulase and cellobiase, and (ii) in second stage glucose is oxidized and converted CO₂ and water.



The microorganisms (e.g., basidiomycetes, clostridium) responsible for degradation of cellulose through the excretion of enzymes (cellulase & Cellobiase)

Protein Decomposition

- Proteins are complex organic substances containing nitrogen, sulphur, and sometimes phosphorus in addition to carbon, hydrogen and oxygen.
- Proteins are first hydrolyzed to a number of intermediate products eg. Proteases, peptides etc. collectively known as polypeptides
- The intermediate products so formed are then hydrolyzed and broken down ultimately to individual amino acids, or ammonia and amides. The process of hydrolysis of proteins to amino acids is known as “ammonification”, brought about by certain enzymes, collectively known as “proteases” or “proteolytic” enzymes
- Amino acids and amines are further decomposed and converted into ammonia.

Lignin Decomposition

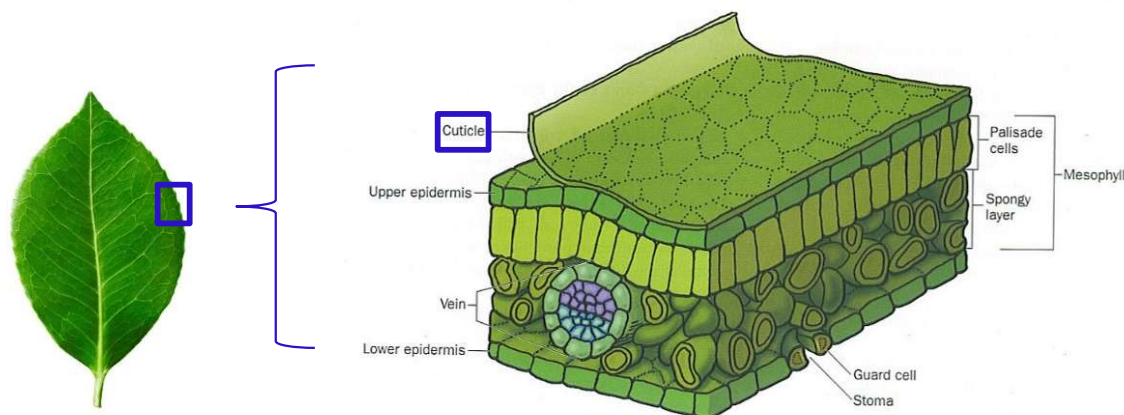
- Lignin is the third most abundant constituent of plant tissues, and accounts about 10-30 percent of the dry matter of mature plant materials.
- Lignin content is one of the most resistant organic substances for the microorganisms to degrade however certain white rot fungi (e.g., Basidiomycetous) are known to degrade lignin at slow rates.
- Complete oxidation of lignin result in the formation of aromatic compounds such as syringaldehydes, vanillin and ferulic acid.
- The final cleavages of these aromatic compounds yield organic acids, carbon dioxide, methane and water.

Lipids decomposition

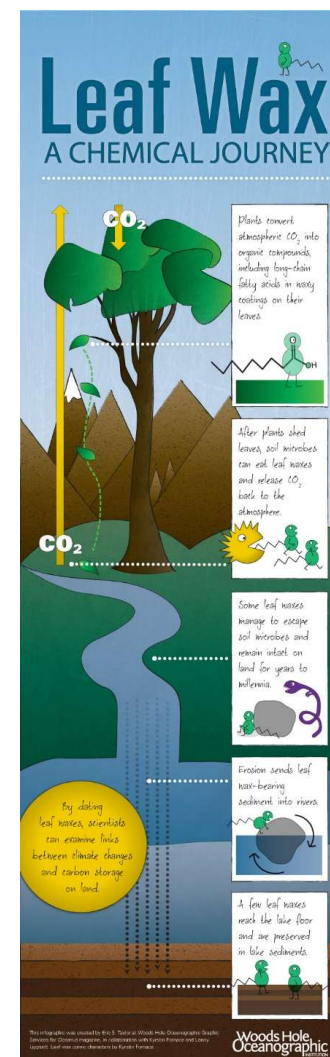
- Lipids are hydrolysed for e.g., wax esters are converted in to fatty acids and alcohols .
- The waxy component are among the best preserved biomacromolecules over geological time period.

Biomarker

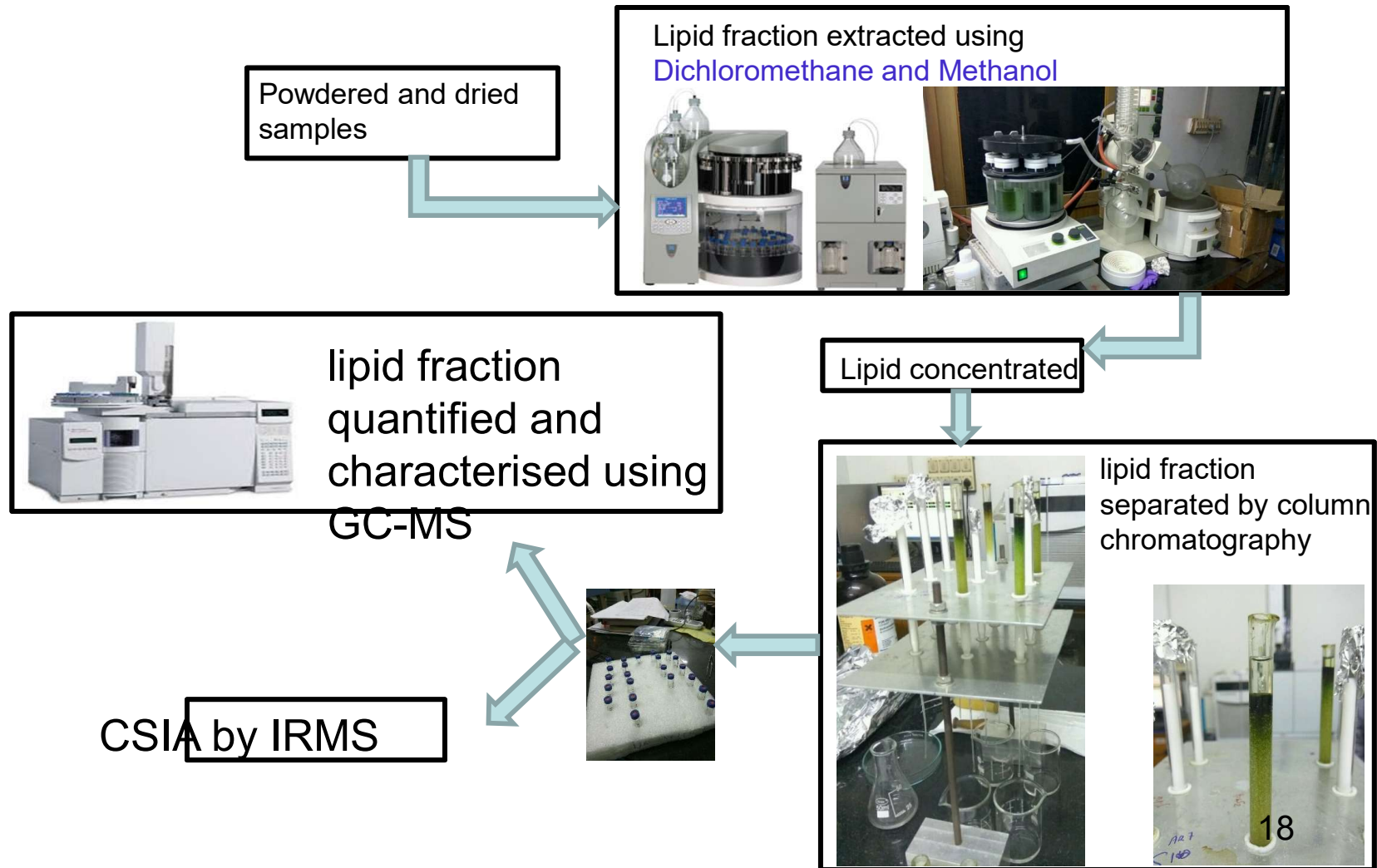
- **Biomarkers:** Compounds that have biotic; source of organic compounds; environmental condition



- **Epicuticular wax:** The lipid components of the protective waxes that coat the leaf surfaces of almost all land plants are environmentally persistent.
- *Common constituents of epicuticular wax are predominantly straight-chain aliphatic hydrocarbon that may be saturated or unsaturated and contain a variety of functional groups.*

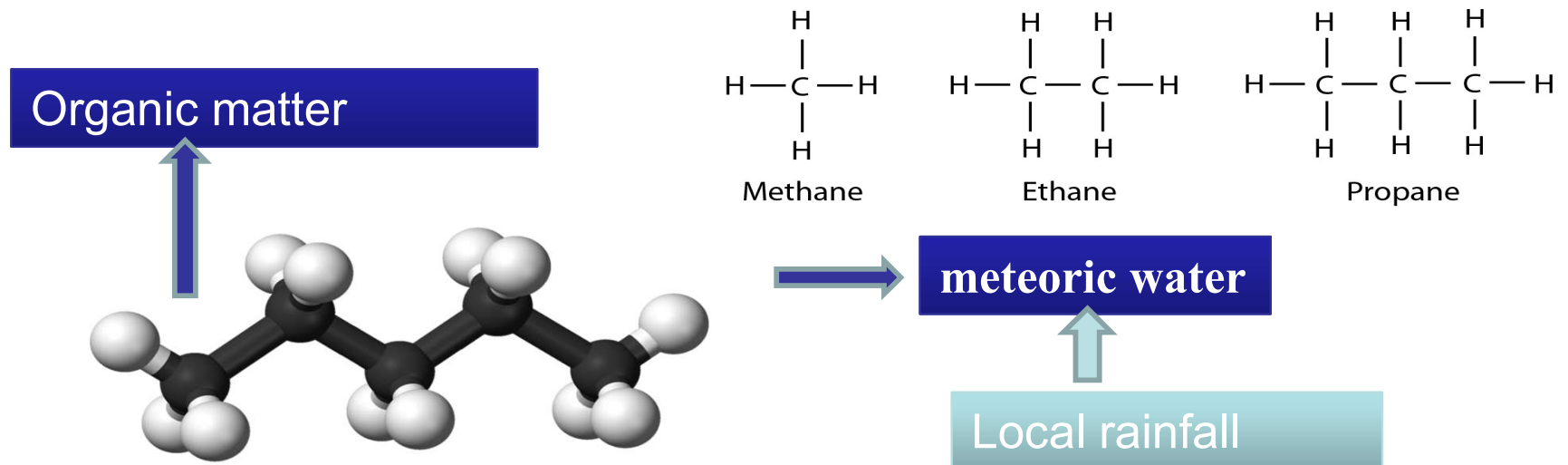


Methodology



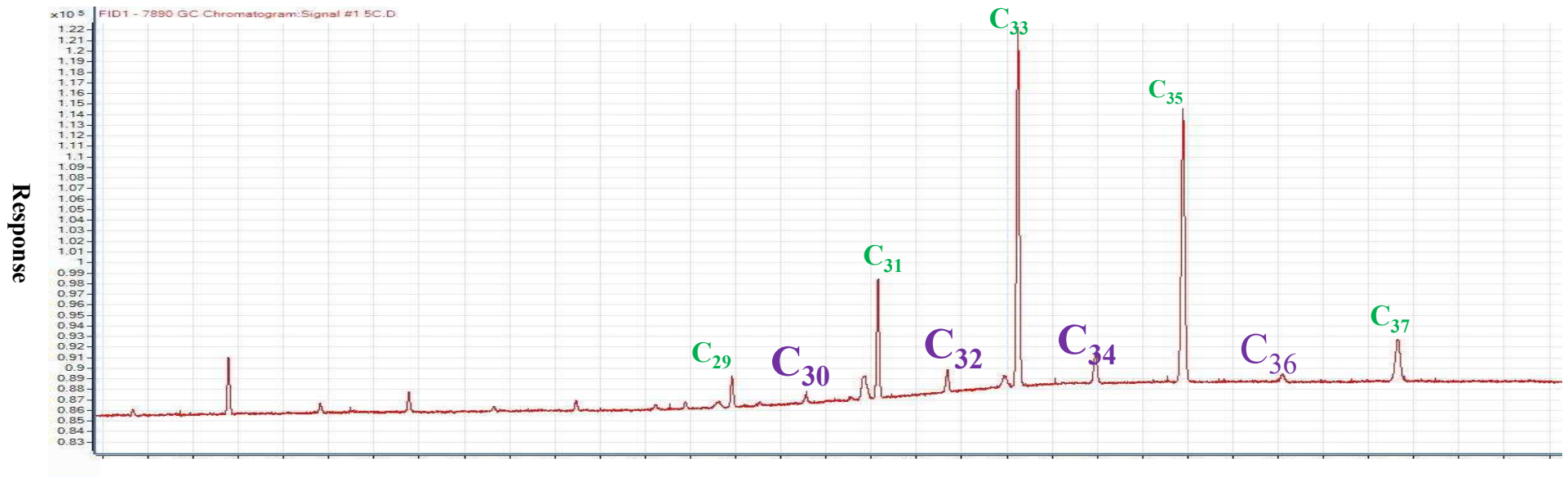
n-alkane

- *n*-Alkanes C_nH_{2n+2} are produced in variable quantities by plants, algae, and bacteria in many environmental settings
- Resistance to degradation and early diagenetic alteration, preserve the history of lipid input to sediments (Giger et al., 1980; Meyers, 1997; Radke et al., 2005).

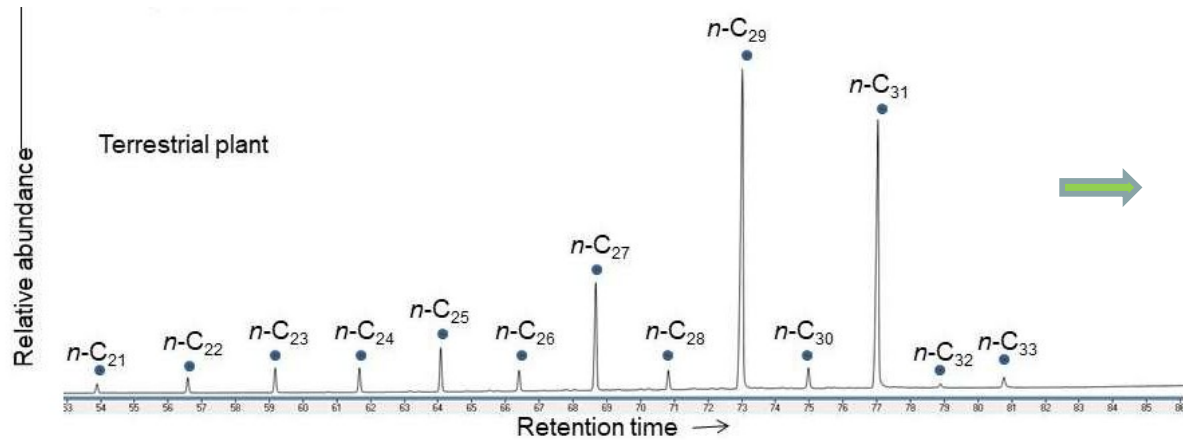


GC-MS chromatogram

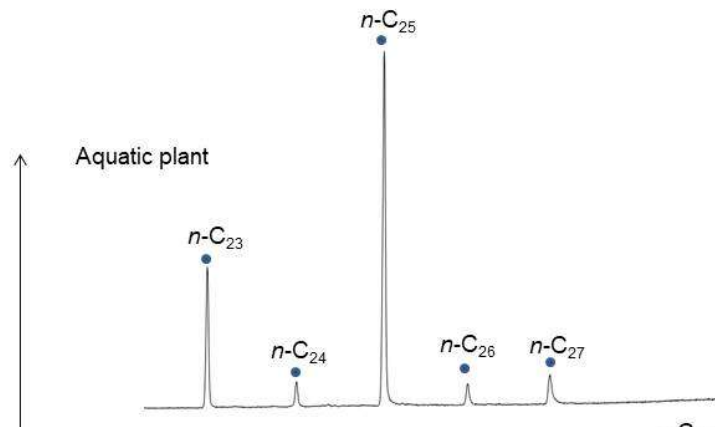
GC-MS (Gas Chromatograph coupled with Mass spectrometer) **chromatogram** of modern leaf



Source determination



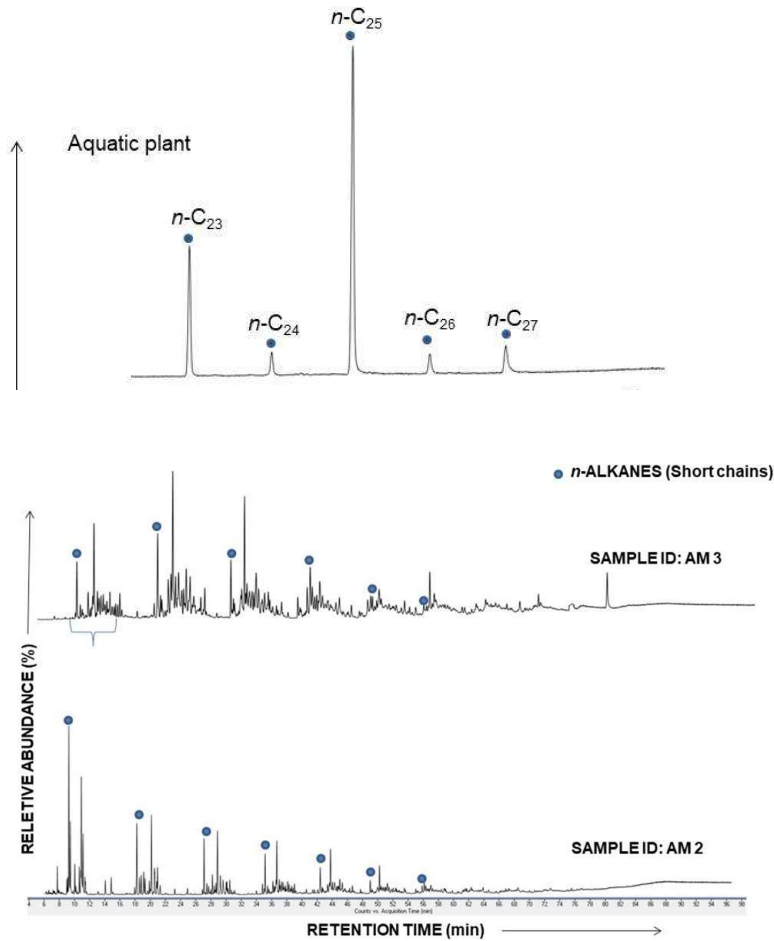
Terrestrial higher plants



Aquatic plants

Behera et al., 2022 (Biogeochemistry)

Source determination



Aquatic plants



Microbial mats

Reference: Behera et al., 2022
(Biogeochemistry)

Proxies based on n-alkanes

$$P_{aq} = (C_{23} + C_{25}) / (C_{23} + C_{25} + C_{29} + C_{31})$$

P_{aq} approximates the proportions of aquatic macrophyte vs. emergent and terrestrial plant inputs to lacustrine sediments. Relatively high values (more than **0.4**) indicate a predominance of aquatic plants (Ficken et al., 2000; Seki et al., 2009).

$$CPI = \text{odd}\Sigma[C_{21-33}] / \text{even}\Sigma[C_{22-32}] \quad \text{Poynter (1990)}$$

The CPI values of immature sediments containing high amounts of terrestrial organic matter lie well above unity (**≤ 5**). Bacteria and algae, in contrast, are characterised by weak odd-over-even predominance and produce low CPI values (**~ 1** , e.g. Cranwell et al. 1987). This makes CPI a widely used proxy for investigating the source of organic matter (e.g., Routh et al. 2007).

Proxies based on n-alkanes

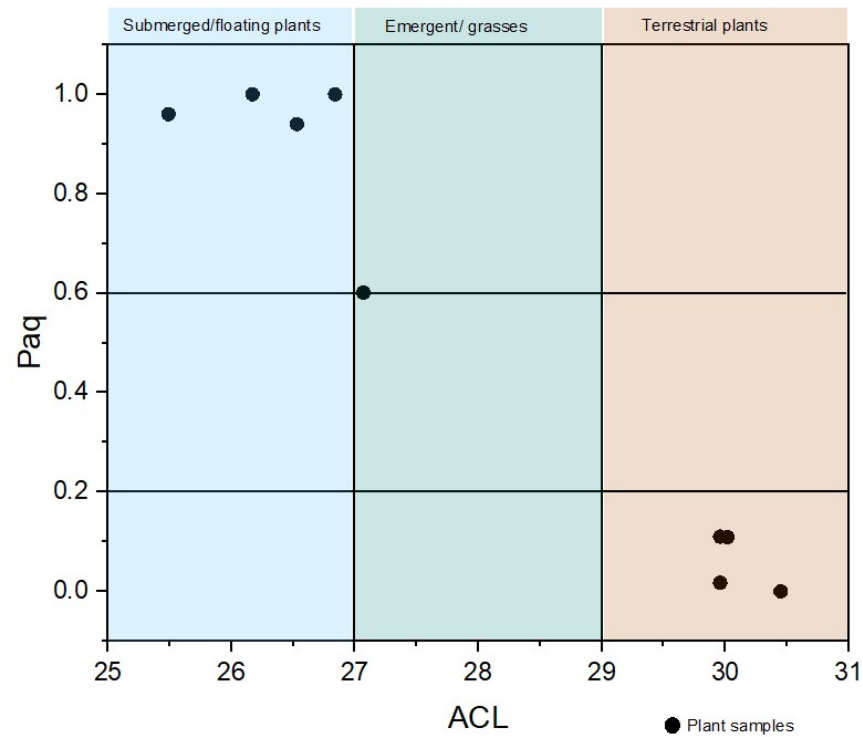
$$\text{Terrigenous/aquatic ratio (TAR)} = (C_{27} + C_{29} + C_{31}) / (C_{15} + C_{17} + C_{19})$$

Higher values for this parameter indicate increased watershed sources of lipid matter relative to aquatic sources.

$$ACL_{27-33} = (27 \times C_{27} + 29 \times C_{29} + 31 \times C_{31} + 33 \times C_{33}) / (C_{27} + C_{29} + C_{31} + C_{33})$$

The ACL index from sediment cores is sometimes applied to detect vegetation or even climate changes over time (Hughen et al., 2004). High ACL values (≤ 29) of leaf wax *n*-alkanes indicates drier environment (e.g., Rommerskirchen et al., 2006)

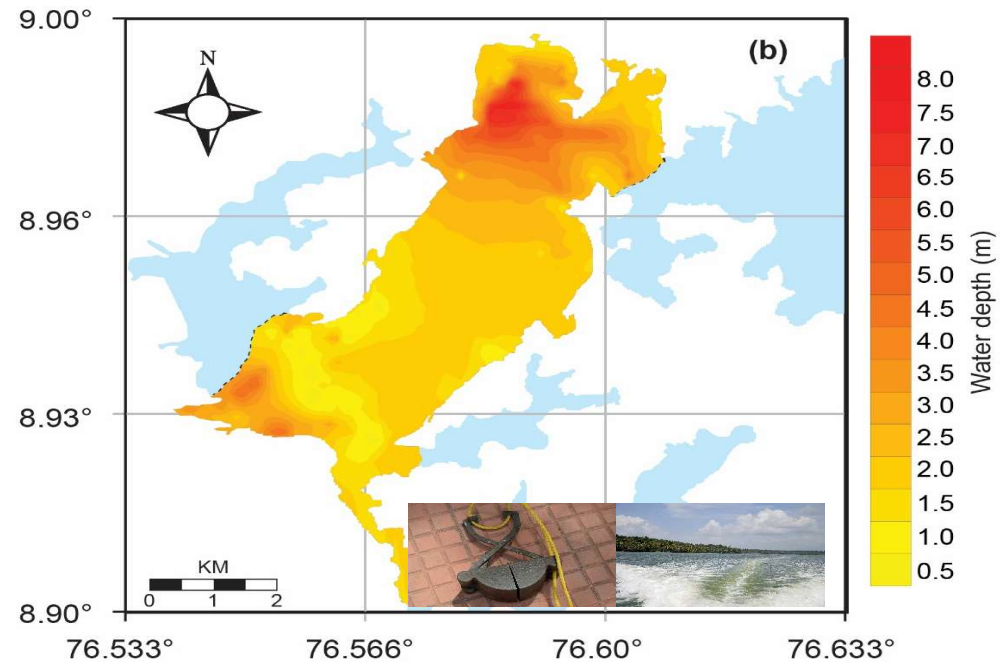
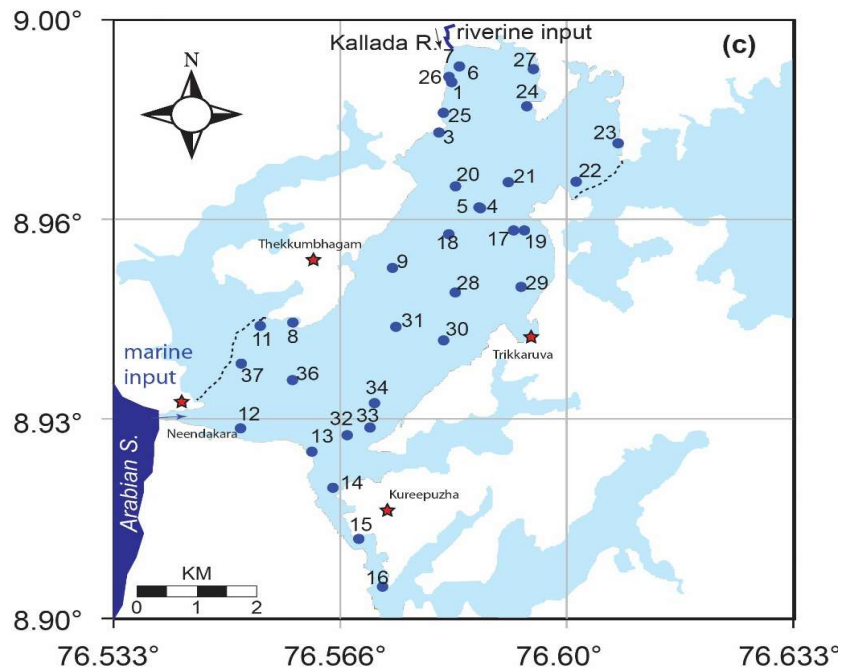
Modern vegetation



Biplot of ACL vs Paq

Case study- Asthamudi Estuary

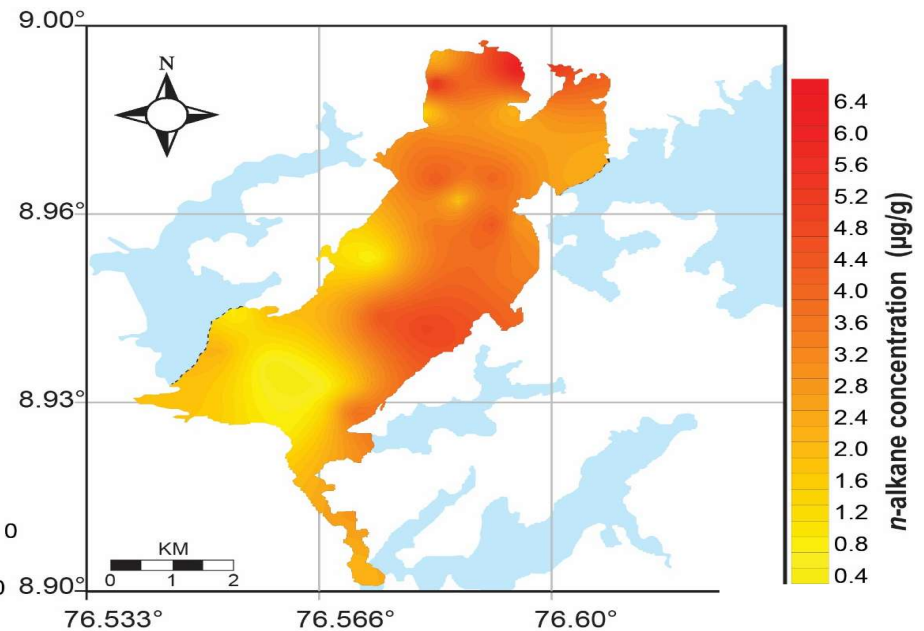
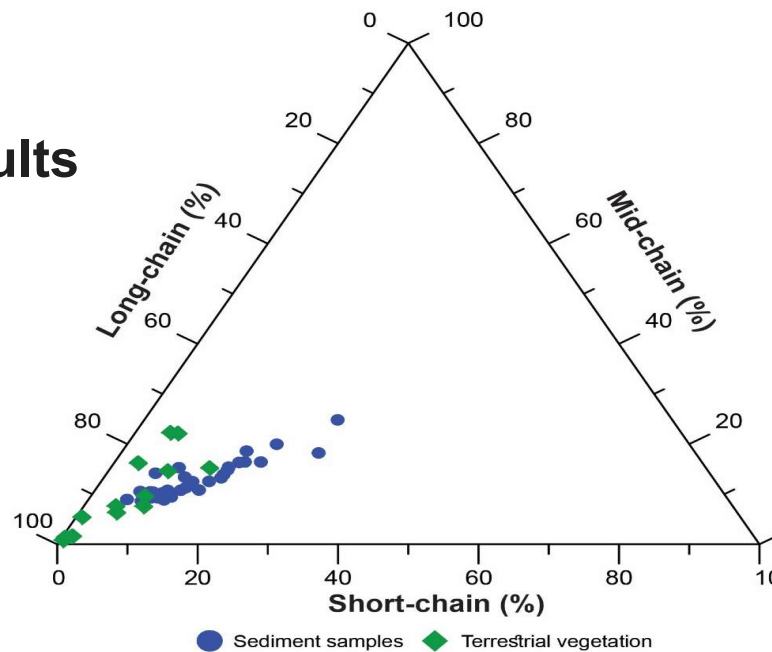
Field sampling



Ankit et al., 2017 (Marine chemistry)

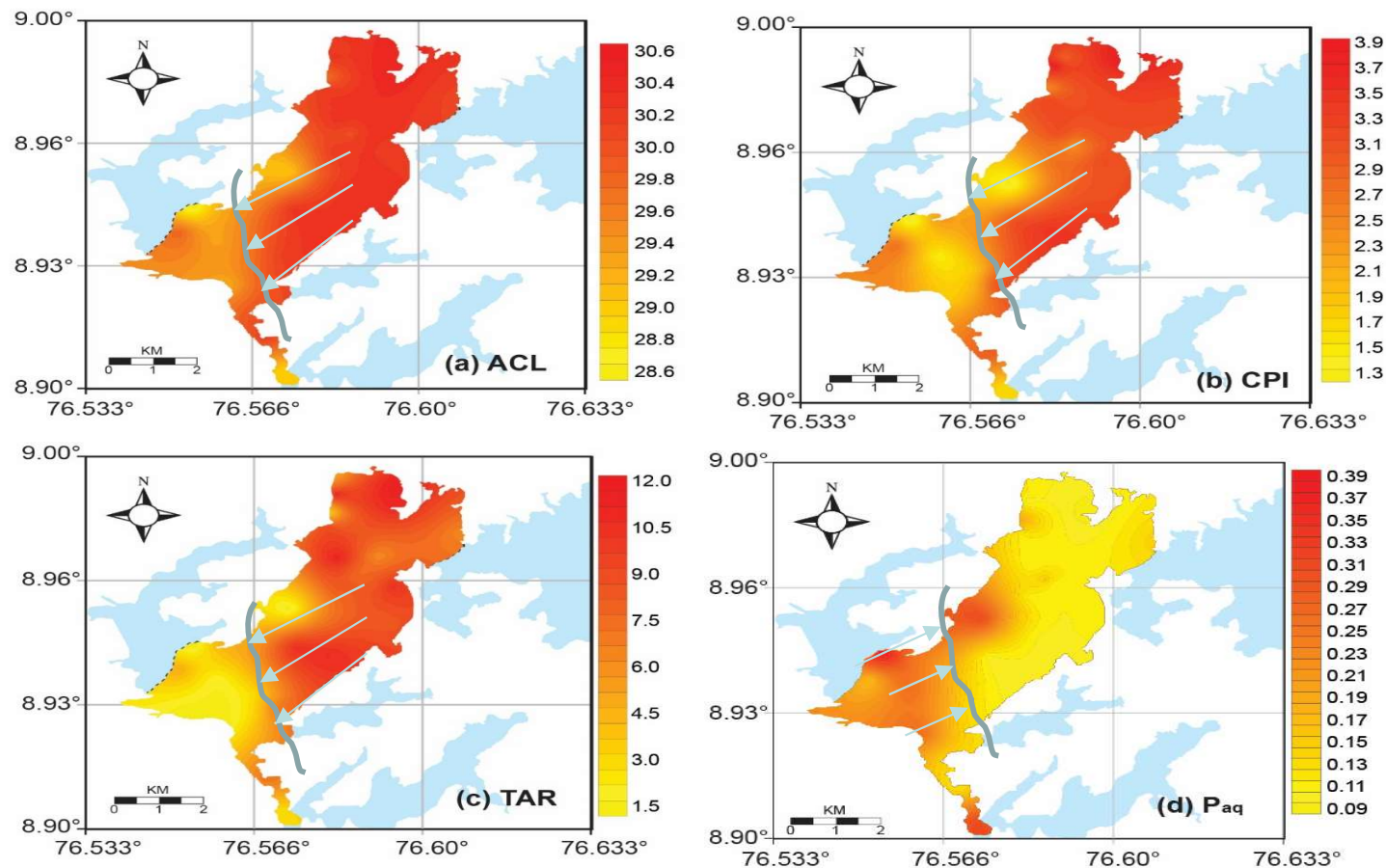
Organic matter Source Determination

Results



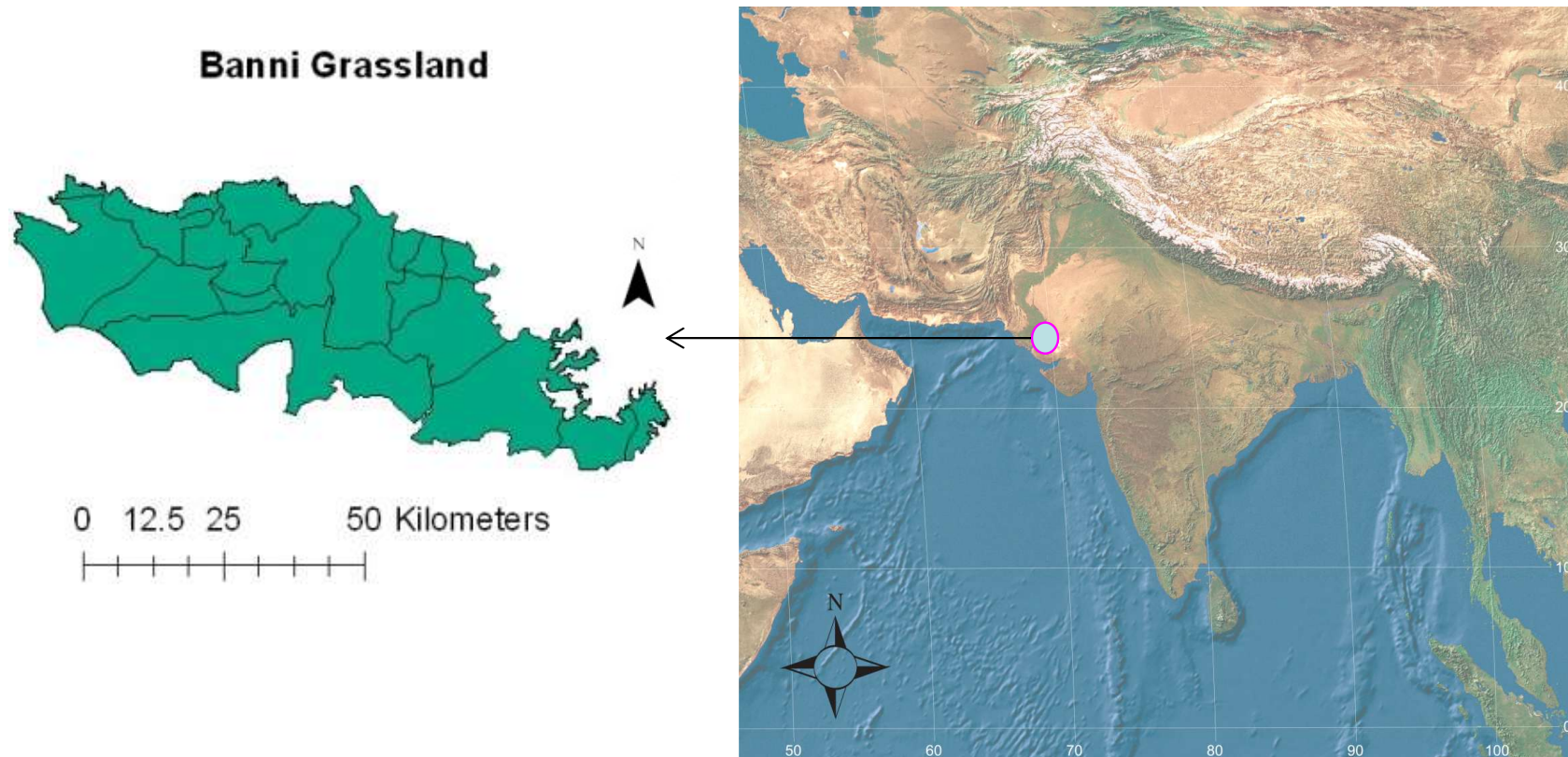
Ankit et al., 2017
(Marine chemistry)

Organic matter Source Determination



Reference: Ankit et al., 2017 (Marine chemistry)

Compound specific isotopic analysis (CSIA)

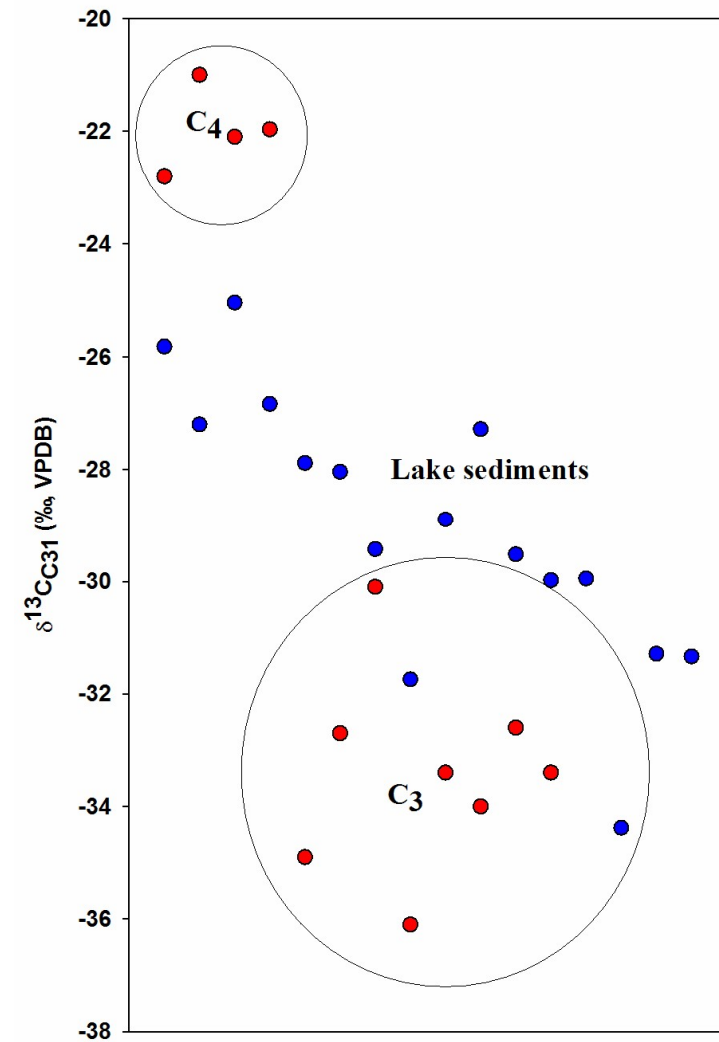


$\delta^{13}\text{C}_{\text{C31}}$ of *n*-alkane

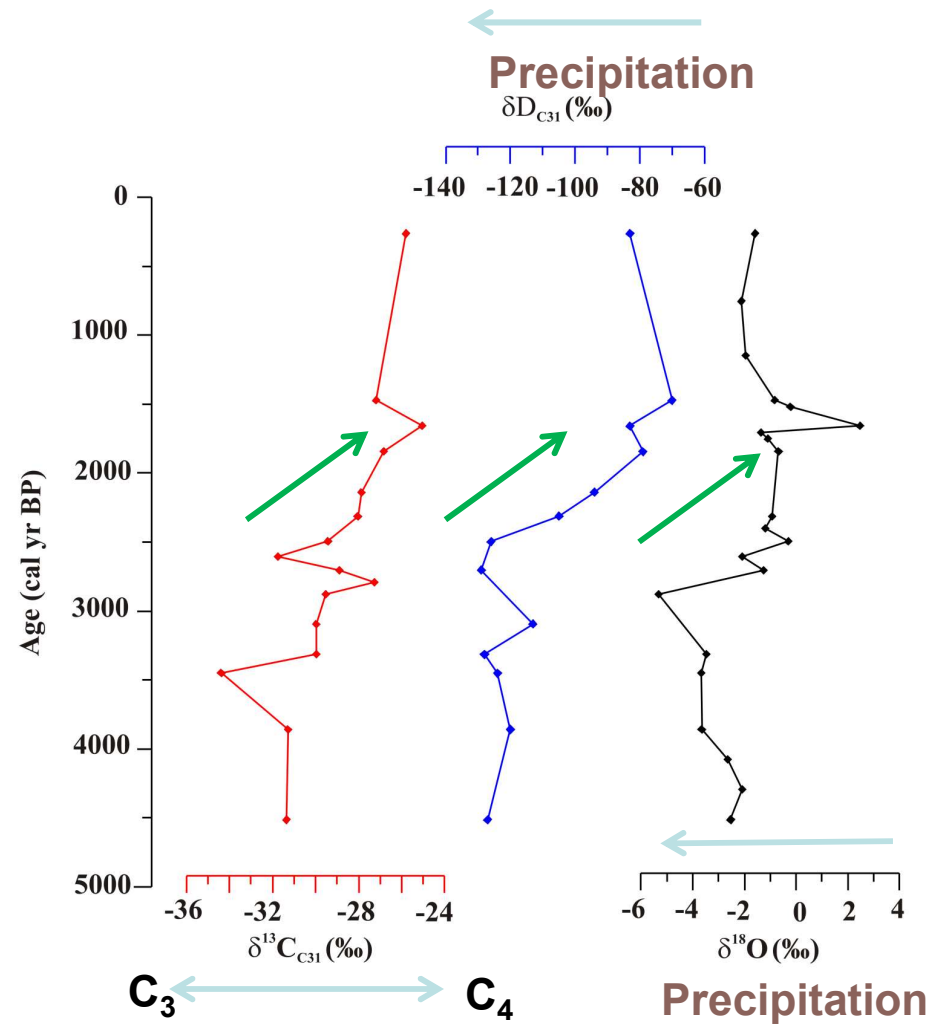
$\delta^{13}\text{C}_{\text{C-31}}$ of modern vegetation

C3 plants = -33.6 ± 1.8 (n=8)

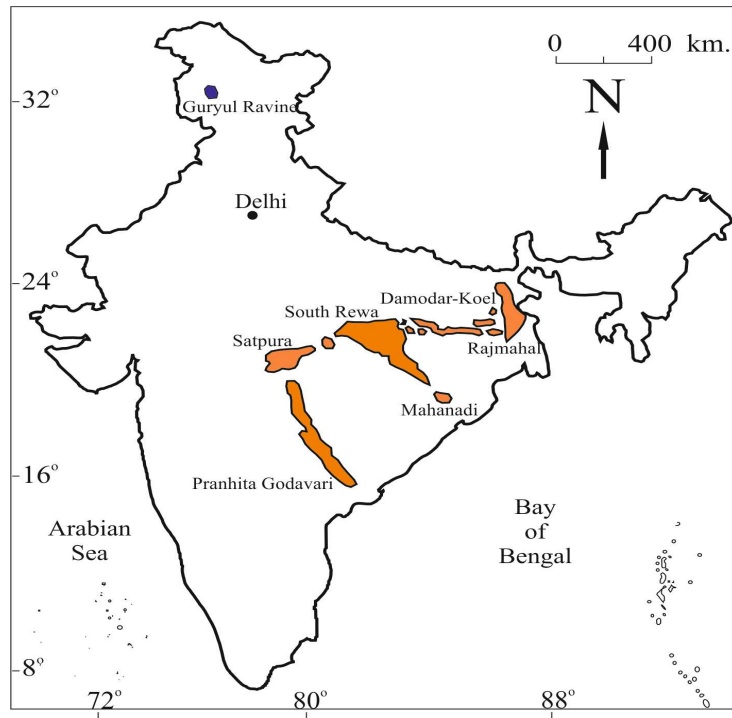
C4 plants = -22 ± 0.9 (n=4)



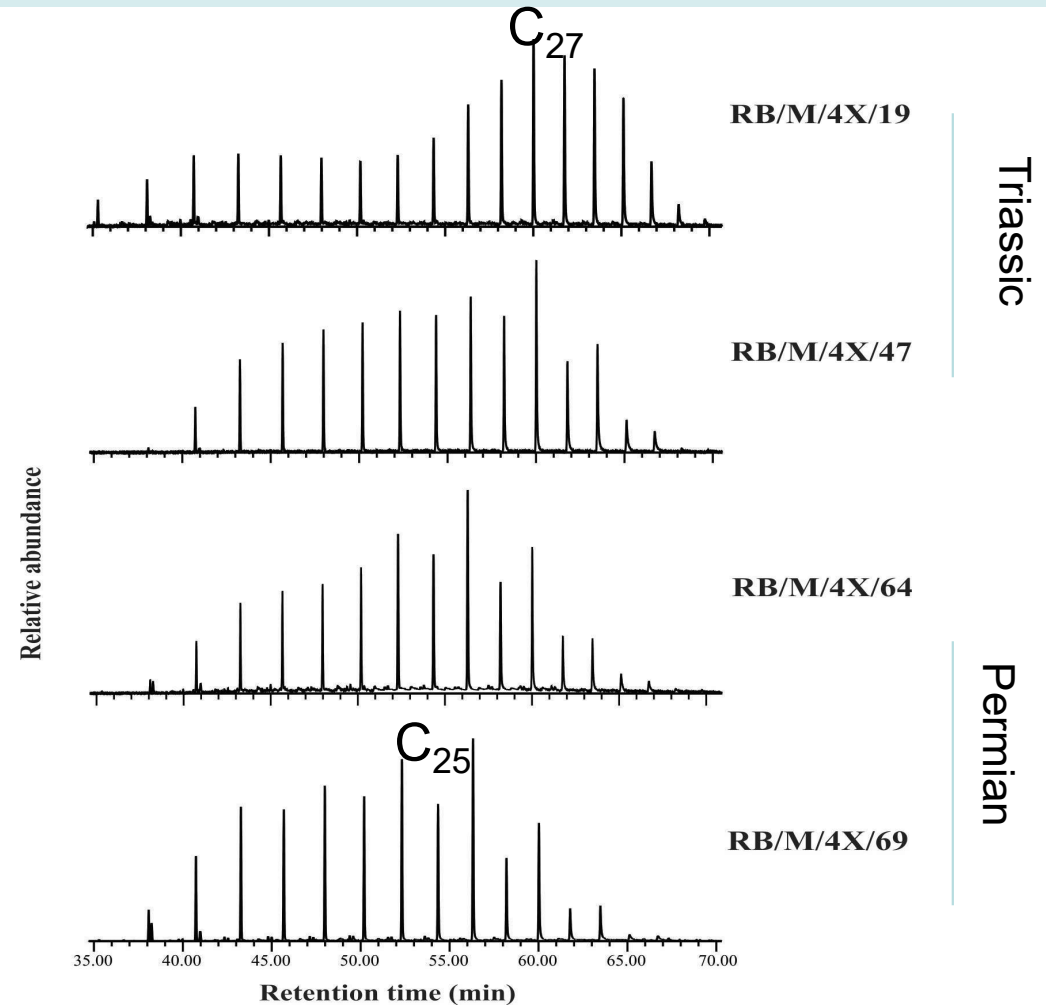
Banni Grassland



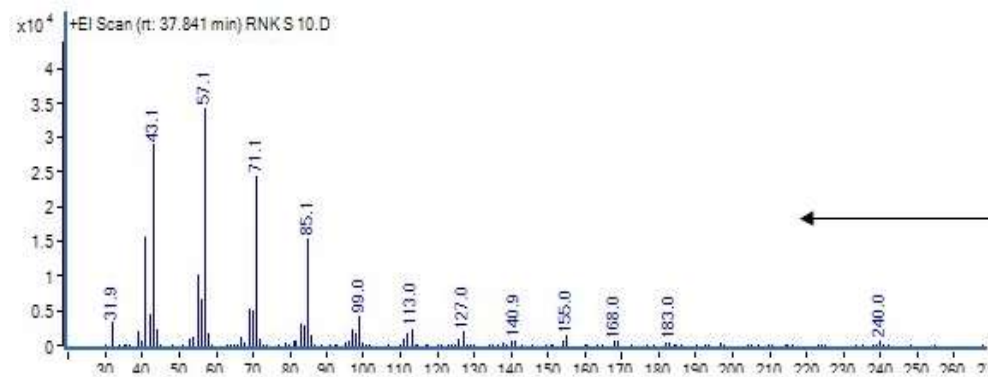
Biomarker tool for long-term vegetational shift



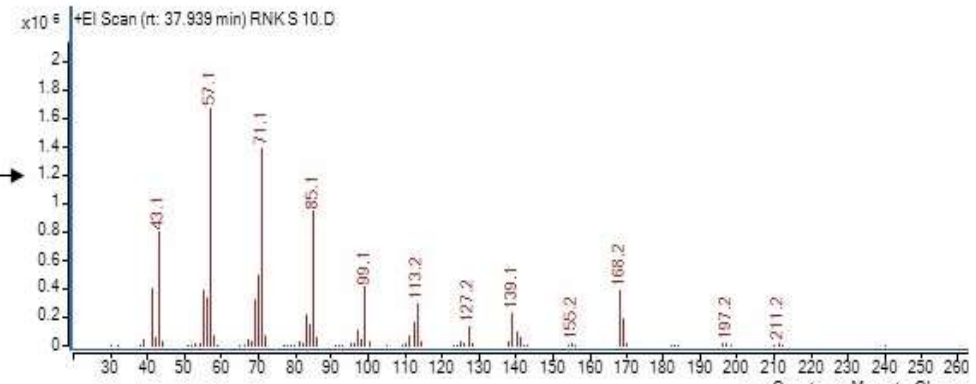
Sedimentary basins in India where continental Permo-Triassic successions are found



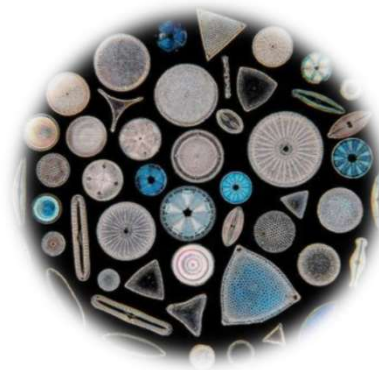
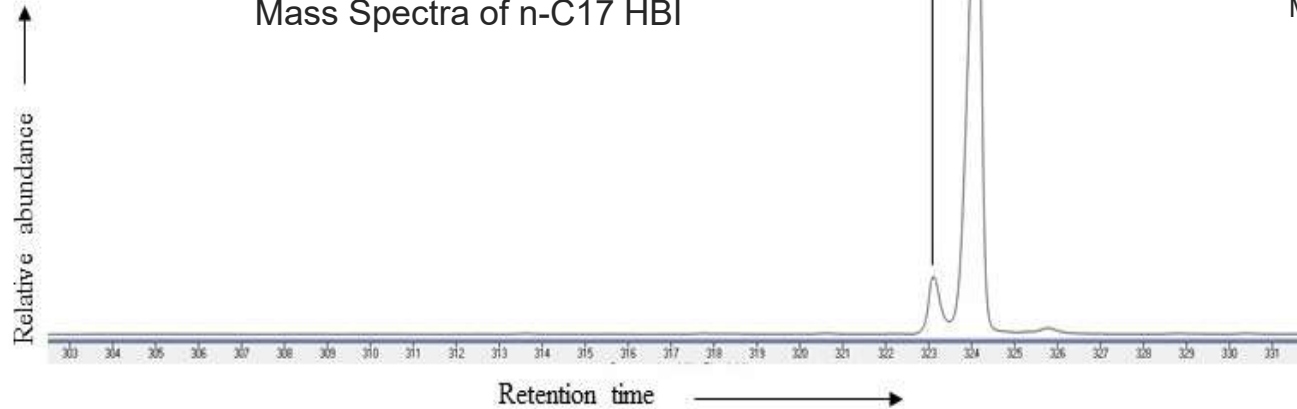
Source specific Biomarkers



Mass Spectra of n-C17 HBI

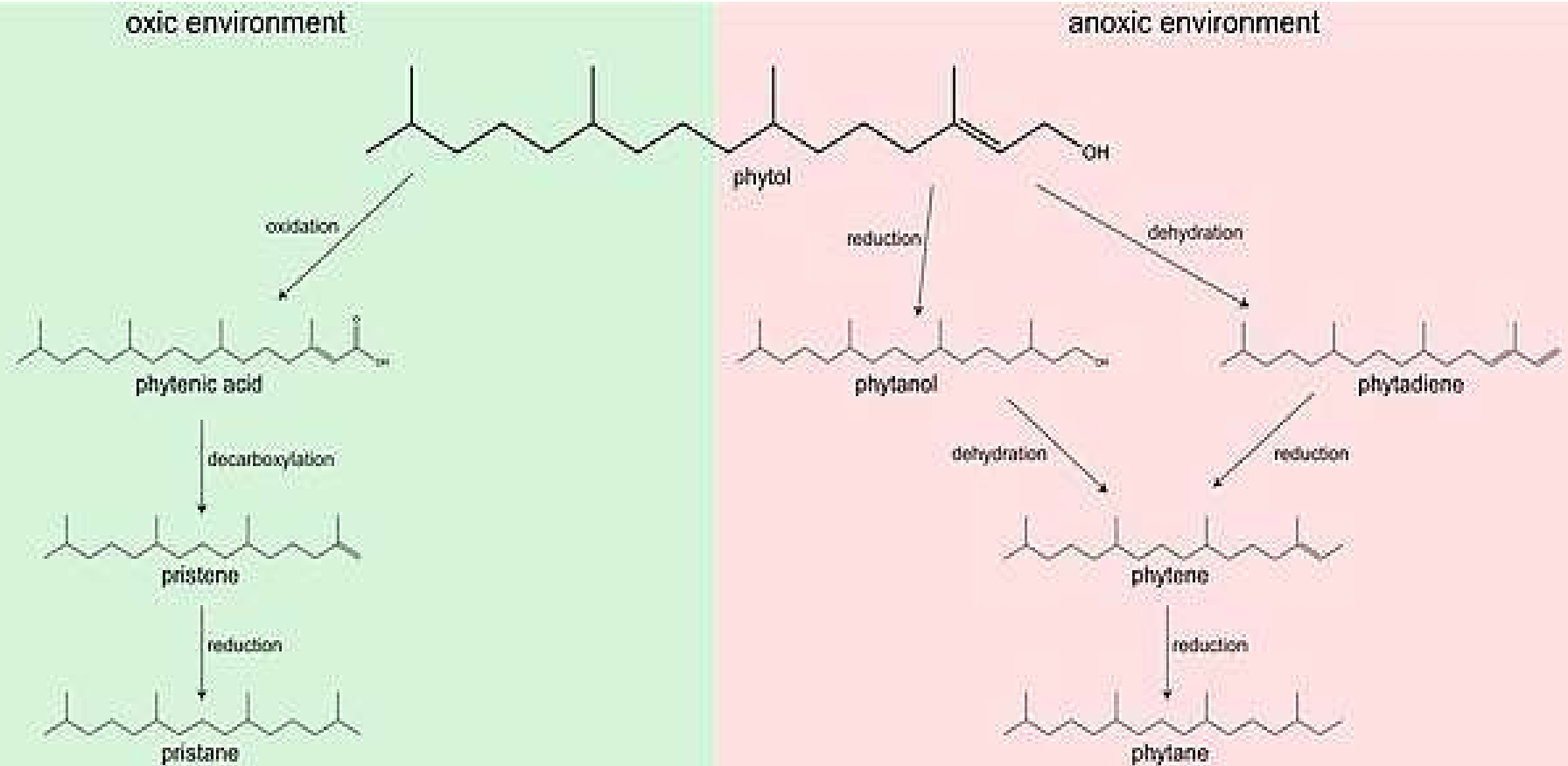


Mass Spectra of C20 HBI

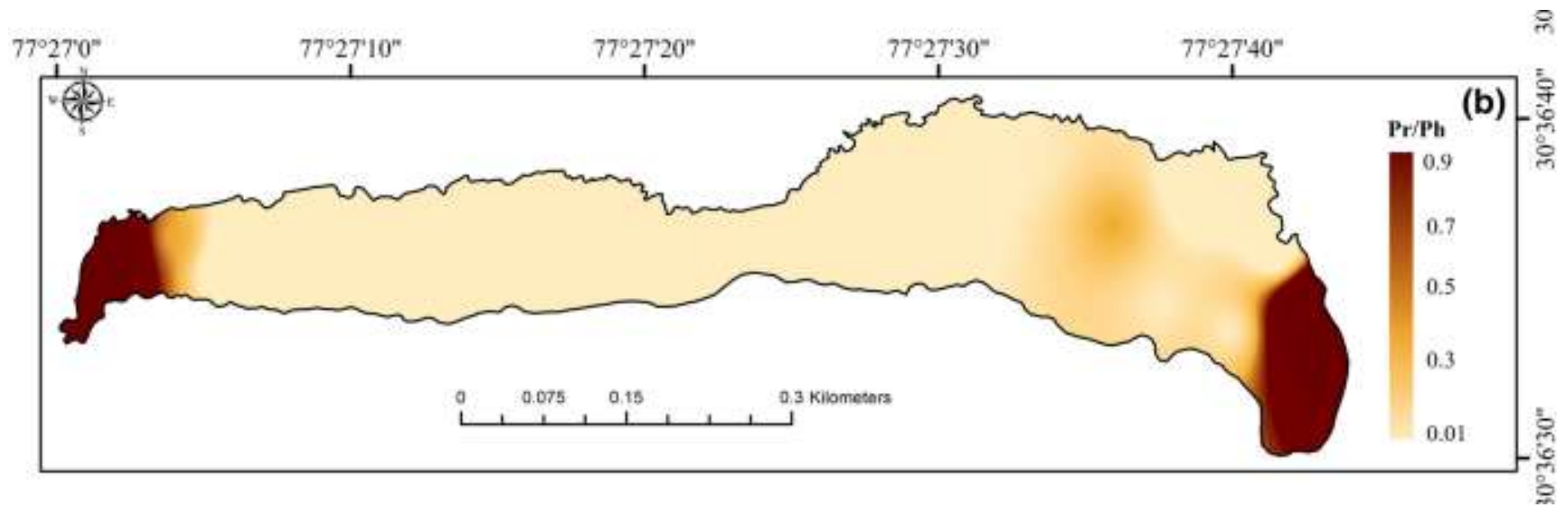


Diatoms Ooze

Implication for depositional environment



Implication for depositional environment



References: Behera et al., 2022
(Biogeochemistry)

Composition of organic matter

Lignin

- Lignin, characterized by phenolic structures (aromatic six-member carbon rings bonded to OH groups)
- Component of the structural tissues of higher (Vascular) plants.
- Lignin forms around the fibrous woody core (xylem) of terrestrial plants and acts as a support structure as the plant grows. Cellulose constitutes up to 40–60% of wood and lignin essentially makes up the remainder.

Lignin

- Lignin constituents are potentially useful indicators of different type of vascular Plant tissues
- Alkaline oxidation of lignin using copper oxide yields four group of structurally related products: p-hydrophenyl, Vanillyl, Syringyl and Cinnamyl groups

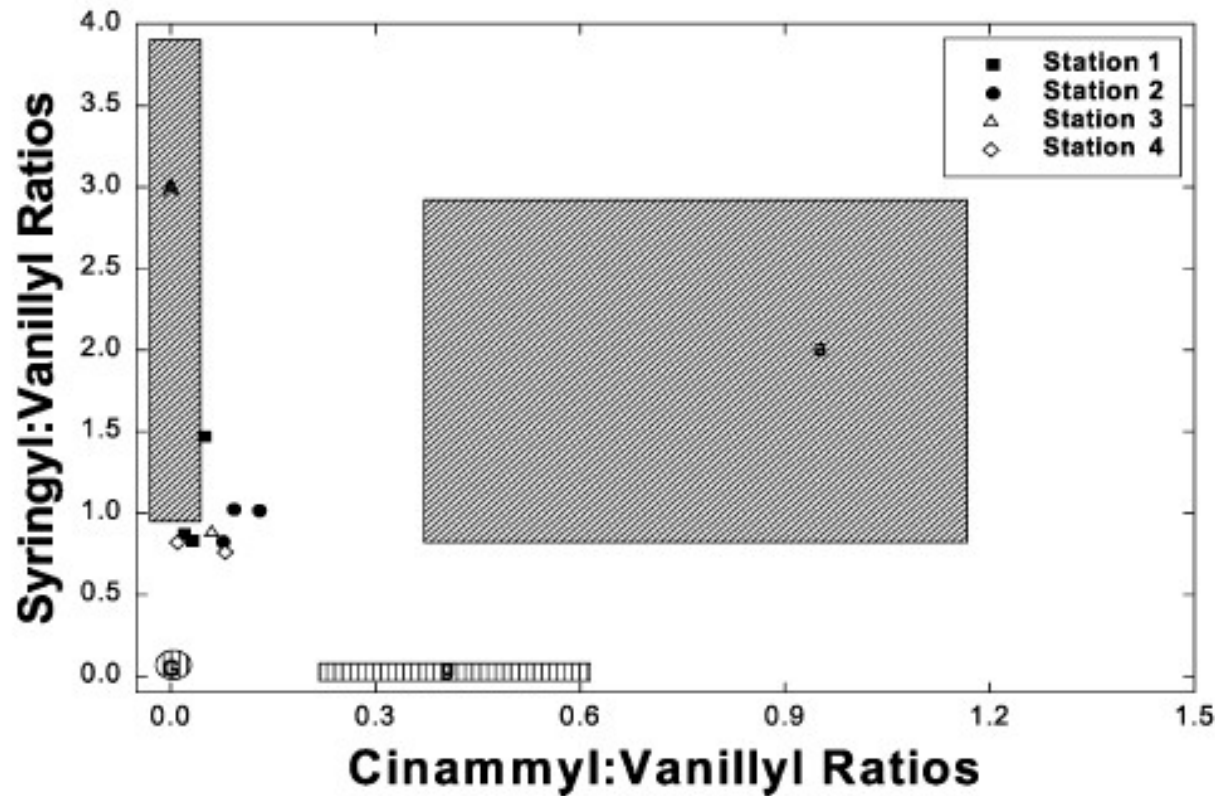
Distinction of Organic matter sources using Lignin

Plant Type	Syringyl (mg/g)	Cinnamyl (mg/g)	Vanillyl (mg/g)	S/V	C/V
Nonvascular plants	0	0	0	— ¹	—
Nonwoody angiosperms	1–3	0.4–3.1	0.6–3.0	0.5–2.5	0.4–1.1
Woody angiosperms	7–18	0	2.7–8.0	0	1.1–5.2
Nonwoody gymnosperms	0	0.8–1.2	1.9–2.1	0	0.4–0.6
Woody gymnosperms	0	0	4–13	0	0

¹—, not applicable.

Lignin oxidation product distribution in different vascular plant tissues (after Hedges and Mann, 1979)

Distinction of Organic matter sources using Lignin



A -Woody angiosperms
a-non-woody angiosperms

G- woody gymnosperms
g. Nonwoody gymnosperms

Station 1 and 2 : Missisipi shelf

Station 2 and 3: Louisiana shelf

Long-term fate of organic matter in the geosphere

3 possibilities

- 1) Exposure to atmospheric oxygen
→ decay, mineralization
- 2) Restricted contact with atmospheric oxygen
→ humification
- 3) Immediate submersion of OM
→ peatification

Long-term fate of organic matter in the geosphere

Diagenesis

- Processes affecting the products of primary production that take place prior to deposition and during early stages of burial
- Relatively low pressure and temperature

Catagenesis

- Thermally mediated alteration of organic matter
- Relatively high pressure and temperature

Long-term fate of organic matter in the geosphere

Maturation

- The evolution of sedimentary organic material with increasing burial and temperature is generally termed maturation